

PHILIP HOLLER

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EDUCATION

Doctoral Candidate in Accounting, University of Mannheim, Germany 2021 – present
Graduate School of Economic and Social Sciences (GESS)
Center for Doctoral Studies in Business (CDSB)
Dissertation: *Essays in the Economics of Decarbonization*
Committee: Stefan Reichelstein and Gunther Glenk (supervisors), Ulrich Wagner (Economics, examiner)
Defended September 25, 2026, with distinction (*summa cum laude*)

Visiting Doctoral Student 01/2026 – 03/2026
Stanford University, Graduate School of Business, California

Visiting Doctoral Student 06/2024 – 08/2024
Université Toulouse Capitole, Toulouse School of Management, France

M.Sc. Management, University of Mannheim, Germany 2018 – 2021
Major in Accounting, Finance, Operations
Semester abroad: ESEADE Buenos Aires, Argentina (2019)

B.Sc. Business Administration, University of Mannheim 2014 – 2017
Semester abroad: UQAM Montréal, Canada (2016)

RESEARCH INTERESTS

- Economics of low-carbon technologies, including electrolytic hydrogen and carbon dioxide removal
- Carbon accounting, emission allocation, and investment incentives
- Cost-efficient carbon removal portfolios and data-center decarbonization

PUBLICATIONS

Glenk, G., **Holler, P.**, Reichelstein, S. (2026). *How carbon accounting rules shape incentives for hydrogen production*. Nature Communications, 17, 7260. doi:10.1038/s41467-026-75473-z.

Glenk, G., **Holler, P.**, Reichelstein, S. (2023). *Advances in power-to-gas technologies: Cost and conversion efficiency*. Energy & Environmental Science, 16, 6058–6070. doi:10.1039/d3ee01208e.

WORKING PAPERS

Holler, P., Reichelstein, S. (2026). *Cost Comparisons for Emerging Carbon Dioxide Removal Technologies*.

Holler, P., Merke, T. (2026). *Dynamic Negative Emissions Portfolios under Paris-Aligned Removal Targets*.

Holler, P. (2026). *A Comparative Rating Framework for Carbon Removal Credits*.

OTHER PROJECTS

Bach, A., Derayati, R., **Holler, P.**, Reichelstein, S. *Optimal Energy Portfolios for Data Centers*. Exploratory case study.

RESEARCH & PROFESSIONAL EXPERIENCE

Staff Researcher 2022–present
Mannheim Institute for Sustainable Energy Studies

- Develop research questions and computational models for assessing the cost and emissions performance of low-carbon hydrogen and carbon dioxide removal technologies.
- Analyze how carbon accounting rules and policy support shape firms' investment and operating decisions.
- Contribute to teaching curriculum, thesis supervision, and organizational activities.

Co-Founder and CEO YAB GmbH Local sustainable consumption and discovery app; SaaS provider.	2021 – 2025
Project Consultant – Indirect Procurement Hilti AG Digital service procurement and sustainable sourcing.	Sep – Dec 2020
Intern – Indirect Procurement Hilti AG Implementation of SAP Ariba as e-procurement tool, sourcing decisions.	Jun – Aug 2019
Intern – Transactions, Advisory PricewaterhouseCoopers Bank valuation projects (peer groups, betas, cost of capital estimation).	Jun – Aug 2016

TEACHING EXPERIENCE

Master Seminar on Corporate Decarbonization (MAN 771)	2022 – 2026
Master Seminar on Corporate Carbon Accounting (ACC 770)	2022 – 2023
ESG Regulation and Sustainability Reporting (ACC 570)	2025, 2026
Supervision of Bachelor and Master Theses	2022 – 2026

RESEARCH SOFTWARE & TOOLS

Low-Carbon Hydrogen Optimization Model

Python/Gurobi model code accompanying the Nature Communications paper, optimizing capacity investments in renewable composition and PtG size, including operational optimization across emission accounting rules

LCOE Calculator (with Thimo Merke and Minghao Chen)

Interactive Shiny for Python application comparing solar and wind generation costs across U.S. states and market segments.

CONFERENCES & WORKSHOPS

TRR 266 Annual Conference – Accounting for Transparency, Mannheim	2023
Mannheim Energy Conference	2023
Hydrogen Colloquium, Fellbach	2023
NHH Bergen Workshop: Corporate Sustainability	2024
IESE Munich: Workshop on Corporate Carbon Accounting, Munich	2024, 2025, 2026
Workshop: Advances in Climate Technologies, Mannheim	2024
Conference on Economics of Energy and Climate, Toulouse Capitole	2024
Accounting Research Camp on Transparency in Corporations and Markets, Obergurgl	2025
Workshop: Carbon Accounting Standards Initiative, Frankfurt	2025
MISES Symposia Presentations (quarterly)	2022 – 2025

SUMMER SCHOOLS

Energy Technology, Policy and Politics Summer School, ETH Zürich	2023
Climate Change AI	2023
H2type. Scientific Exploration of the Green Hydrogen Future, TU Dresden	2024

GRANTS & SCHOLARSHIPS

PhD Scholarship, GESS	2021 – 2022
IDEUM Scholarship for internationalization of doctoral education	2023
ENGAGE.EU Mobility Grant	2024
ENGAGE.EU Workshop Funding for Early Stage Researchers	2024
TRR Mobility Funding for PhD	2025

VOLUNTARY ACTIVITIES

Teaching, various sustainability projects (Peru, Chile, Bolivia, Argentina)	2018
Team Lead - Culture, VISUM e.V. Mannheim	2020
CFO, ESN VISUM Mannheim e.V.	2021 – 2022
Member of the Supervisory Board, ESN VISUM Mannheim e.V.	2022 – 2023
Kinderuni (Children's University Mannheim)	2024
(Co-)Reviewer for Nature Energy, Nature Communications, Energy & Environmental Science, Renewable and Sustainable Energy Reviews	

LANGUAGES

German (native), English (C2), French (C2), Spanish (C1)

TECHNICAL SKILLS & TOOLS

Programming and analysis: Python (analytics, ML, modeling, optimization), Gurobi, R, Stata, MATLAB.
Energy modeling: Velocity Suite, SAM, GREET.
Writing and productivity: L^AT_EX, MS Office.